

A LECTURE ASIDE

What's on the shirt

A theoretical model, and how one fits on a t-shirt

The compact CERN form. The long typeset version comes from Veltman's Diagrammatica, transcribed by T. D. Gutierrez.



THIS IS STILL A COURSE ABOUT MODELS

A theoretical model

Some of our models are built from data, like a histogram. Some generate data, like your random number generators. This one is a third kind: what we know about the world, written down as mathematics.

1

Observe

Measurements, regularities, things that always happen the same way.

2

Encode

Write the regularity as mathematics — the smallest statement that contains it.

3

Generate

Turn the crank. The mathematics hands back motion, rates, spectra.

4

Compare

Check the prediction against a measurement. It lands where predicted: the model is right. Anywhere else: it is wrong.

Not meant as an approximation of reality, but as our best statement of what reality is.

WHAT'S ON THE SHIRT

Why it's called a Lagrangian

- Newton gave us forces, one situation at a time.
- Underneath is one function, the Lagrangian, and one principle: of all possible paths, a system takes the one that makes the action stationary.
- Quantum mechanics: add up all the paths and you get probabilities.
- The same machinery works for the classical and the quantum world. It fails only for quantum gravity.

ONE FUNCTION IN

$\mathcal{L}$



ALL THE DYNAMICS OUT

equations of motion · conservation laws · interaction rates
· decay probabilities

The shirt isn't a summary of the Standard Model. It is the Standard Model: everything we know except gravity.

Four ideas, one language

1

Newtonian mechanics

Bodies with definite positions and momenta. Know the parameters, know where it goes.

2

Relativity

Go fast and adding velocities breaks. The real fact is a speed limit built into spacetime.

3

Fields

Electricity and magnetism kill action-at-a-distance. Charges talk to a field — and it moves at that same limit.

4

Quantum mechanics

Go small and you cannot pin the parameters down. You only predict probabilities.

Put all four together and you get quantum field theory: fields are fundamental, particles are their excitations, and you compute probabilities.

STEP 1

Start with the matter fields

- Write down the matter and nothing else.
- Three generations, each with quarks and leptons.
- Massless for now — just propagating.
- This is the ψ on the shirt, plus a derivative for how it moves.

I	II	III
QUARKS u d	QUARKS c s	QUARKS t b
LEPTONS e ν_e	LEPTONS $\mu \nu_\mu$	LEPTONS $\tau \nu_\tau$

Everything we call matter sits in this table.

STEP 2

Demand symmetry — the forces drop out

Insist the physics doesn't change if you relabel a particle's internal phase differently at every point in spacetime. A plain derivative notices. To fix it you must introduce a compensating field.

U(1)

→ the photon

Electromagnetism

SU(2)

→ the W and Z

The weak force

SU(3)

→ the gluons

The strong force

You don't add the forces. They're the price of the symmetry.

THE CATCH

Those symmetries forbid mass

$$m \bar{\psi} \psi$$

not allowed

Write a mass term by hand and the gauge symmetry you just used to generate the forces is broken. The whole construction falls over.

But the world plainly has mass in it.

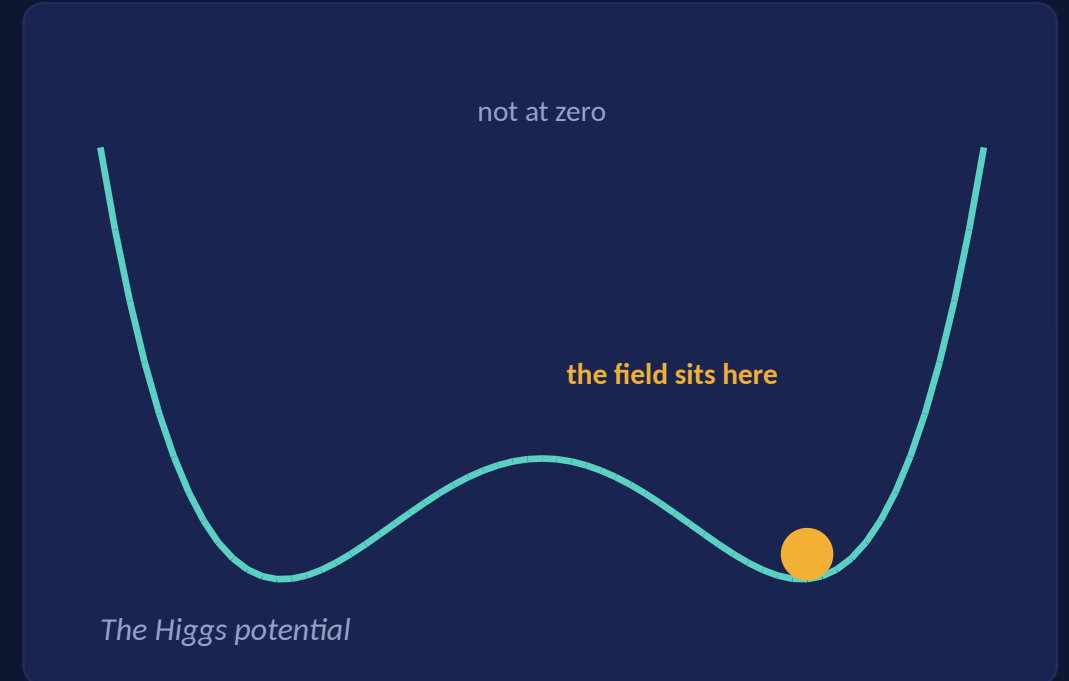
So mass can't be something we write into the theory. It has to be something the particles acquire — from somewhere else, by a mechanism the symmetry tolerates.

That is the entire reason the next slide exists.

STEP 3

Add one more field, everywhere

- A field that fills all of space.
- It sits not at zero, but at the bottom of its potential — a value the universe settled into early on.
- Anything that couples to that background picks up a drag that behaves exactly like mass.
- The same coupling mixes the quark generations — which is how one family turns into another.



The Higgs field was switched on at the beginning and left on. Mass is what it feels like to be stuck to it.

PUTTING IT TOGETHER

Reading the shirt, term by term

1

ψ

The matter fields

Quarks and leptons, three generations.

2

∂_μ

The derivative

How those fields move through spacetime.

3

D_μ

The covariant derivative

Symmetry imposed — which forces in new fields.

4

$F_{\mu\nu}$

The kinetic terms

Those new fields get a life of their own: the forces.

5

$\phi, V(\phi)$

The Higgs and its potential

Mass, and the mixing between generations.

Symmetry fixes the form of every term. The numbers inside them are not on the shirt.

WHAT THE SHIRT DOESN'T SHOW

19 numbers you have to measure

19

free parameters

The theory can't predict them. Every one had to be measured.

- 6 quark masses
- 3 charged-lepton masses
- 3 force strengths
- 4 quark-mixing (CKM) parameters
- 2 Higgs: background value and mass
- 1 strong-force vacuum angle

The shirt is the model. The 19 numbers are its fitted parameters.

HOW GOOD IS IT?

Right about 5% of the universe

5%

ordinary matter

Everything the Standard Model describes: every atom, star,
planet and person.

With its 19 numbers measured, it gets every experiment on ordinary matter right, often to the limit of our precision.

The rest is about 27% dark matter and 68% dark energy. We don't know what either one is.

And gravity isn't in the Lagrangian at all.

Complete for the 5% we understand. Silent about the rest.

THE HIGGS PROBLEM

A theory in hand. Now what?

- Just after Higgs' paper: you want to find the Higgs boson.
- It decays almost instantly, so you only see what it decays into, for example $H \rightarrow \gamma\gamma$.
- To build the detector you must know: what photon energies does it have to handle?
- The theory only gives probabilities. Each collision is one random outcome, so the answer is a distribution.

ONE THEORY IN

$\mathcal{L}$



MANY RANDOM EVENTS OUT

draw · decay · boost · histogram

Monte Carlo simulation: generate events by the theory's rules, then look at what comes out.

From the Lagrangian to your generator

Generate thousands of Higgs decays at random, each following the theory's rules, and histogram the photons.

1

Draw

Draw each Higgs' speed from a distribution. Your lab's accept-reject can do any distribution.

2

Decay

Two-body decay in the rest frame. The kinematics are fixed.

3

Boost

Lorentz-boost the two photons into the lab frame.

4

Look

Histogram the photon energy spectrum. That's your design spec.

Random variables, distributions, four-vectors, objects — all of it earning its keep on a real question.



Fields, gauge symmetries, the Higgs, and 19 measured numbers: a theoretical model that fits on a t-shirt and gets the 5% of the universe we understand right.

To see what it predicts, simulate it. That's the rest of this lecture.